

1 Sharing sunlight between trackers and crops: dynamic
2 light-sharing control for agrivoltaics under a
3 food-security floor on the world's staple croplands

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6 **Abstract**

Agrivoltaic systems grow crops under photovoltaic (PV) modules, but the electricity they can produce without compromising food production is usually set by a fixed array geometry. Single-axis trackers offer another lever: the daily rotation can be shifted from maximum electricity towards letting light through, a choice we call light sharing. We quantify how much this lever is worth under a food-security floor (90% of the open-field yield) and how robust the answer is. We built a differentiable simulator that couples solar geometry, trackers with backtracking, PV output, crop light and microclimate, a soil-water balance and the SIMPLE crop model with a flowering-stage shade penalty. The PV part agrees with PVGIS at 29 sites ($r = 0.97$, mean bias -4.9%). Because the crop response to shade is the least certain part, we fitted it by gradient descent through the simulator to a meta-analysis and four field trials, obtaining an ensemble of plausible responses; the proportional response used in the main analysis is conservative against the trials, so the reported gains are lower bounds. All methods were compared in a deployable protocol: the array density of every site was chosen from its historical seasons with one rule, and performance was measured on geographically held-out sites and years, for 400 crop-site pairs representing the harvested area of wheat, rice, maize, soybean and potato, in baseline and $+1.5$, $+2$ and $+3$ °C climates. At equal mean yield retention (about 0.905), simple light-sharing rules allowed arrays about three times as dense as static arrays and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24 and electricity from 217 to 387–465 MWh ha⁻¹ yr⁻¹. A rule that follows crop phenology, which needs no weather forecast, was the best hand-written rule, and its gain over static arrays rose from 0.17 to 0.19 and 0.30 LER under calibrated and field-

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fitted responses. A neural controller trained through the simulator added only 0.01–0.05 LER; training it on the ensemble of responses, or giving it a farm-measurable biomass signal, changed nothing, and a feedback rule on that signal was worse than the phenology rule. The decision that mattered was the design: choosing the density under the ensemble mean met the floor at 55–58% of sites when the true response was another plausible one, against 74–76% with an oracle, and choosing it so that the floor holds in 75% of plausible responses restored the oracle level for 0.02–0.03 LER. The floor was met in 70% of individual seasons, and production-weighted retention was below 0.9 for soybean and maize, so per-site enforcement needs region- and crop-specific design. Equipping 5% of the world’s staple cropland at the floor would yield about 14 PWhyr^{−1} but would need about 10 TWp of modules, 5.7 times the installed global PV capacity.

7 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,
8 Crop modelling, Differentiable simulation

9 1. Introduction

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017), and a new generation of climate and
17 crop models shows these impacts emerging earlier than previously expected
18 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
19 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
20 rural electricity demand, so converting farmland into solar farms is tempting
21 and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-

climate. It lowers canopy temperature and evaporative demand, conserves soil moisture and improves water-use efficiency, especially in drylands (Marrou et al., 2013b; Hassanpour Adeg et al., 2018; Barron-Gafford et al., 2019; Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candidate adaptation measure against heat and drought stress, in addition to a mitigation measure through renewable electricity. The same shade, however, also reduces the light available for photosynthesis. Regulators increasingly require AV systems to retain most of the reference crop yield. The German pre-standard DIN SPEC 91434, for example, requires at least 66% of the reference yield (DIN, 2021), and other schemes and many farmers set stricter floors. How much electricity can be produced without sacrificing food security depends on the crop, the local climate, the year and the growth stage.

Most AV designs fix the geometry of the array, i.e. the ground coverage ratio (GCR, module area per ground area) and the tilt or tracking rule, from generic rules of thumb. Single-axis trackers add a degree of freedom: the modules can be rotated away from the sun to let light through when the crop needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the sun to maximise electricity. Choosing the rotation every day is a sequential decision problem. The value of light depends on the crop’s phenological stage, soil water and upcoming weather, and the effect of a decision appears in the harvest months later. Recent work in this journal has used evolutionary multi-objective optimisation to lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for crop management has mainly used reinforcement learning with black-box simulators (Gautron et al., 2022), which is sample-inefficient and hard to scale to global, multi-decadal assessments.

Two questions follow. First, how much of the electricity that an agrivoltaic system can produce under a food-security floor comes from the array density, and how much from dynamic light-sharing control of the trackers? Second, how robust is that answer to the weakest part of any such study, the crop response to shade, and does a controller learned from a simulator add anything to a well-designed hand-written rule? Answering them requires a simulator that couples the physics of light, PV, microclimate and crop growth, a protocol in which all controllers face the same design constraints, a check of the shade response against field data, and enough sites and seasons to separate real effects from noise.

This paper makes three contributions. The first is a quantification of light-sharing control under a yield floor. Light sharing is parameterised by one daily level between full backtracking and the rotation that minimises the

shadow on the crop, and we compare static arrays, a seasonal schedule, a rule that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds and three spatial splits. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS, and the crop response to shade is compared with a meta-analysis (Laub et al., 2022) and with field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023).

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry about three times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The simulated shade response is conservative against published field trials. When it is calibrated to a meta-analysis or fitted to the trials, the allowed density and the gain of light sharing rise, and the ranking of methods is unchanged (Section 3.6). The gains reported here are therefore lower bounds with respect to the trials we could use.
4. A learned controller adds a smaller gain over the best rule, 0.01–0.05 LER (about 10% more electricity at equal retention in the main analysis). Training it on an ensemble of shade responses, or giving it a

- 104 farm-measurable biomass signal, does not change this (Section 3.7).
- 105 5. The bottleneck is the uncertainty of the crop response that enters the
 106 design. Gradient-based calibration through the differentiable simulator
 107 gives an ensemble of plausible responses, and a density chosen so that
 108 the floor holds in 75% of them restores the compliance of an oracle
 109 design (74–76% of sites against 55–58% for a design under the mean
 110 response) for 0.02–0.03 LER (Section 3.7).
- 111 6. At the yield floor, agrivoltaics on 5% of the world’s staple cropland
 112 could supply about 14 PWh yr^{−1}, but only with about 10 TWp of mod-
 113 ules, several times the installed PV capacity of the world (Section 3.9).

114 2. Materials and methods

115 2.1. Global cropland sites, weather and crop calendars

116 We studied five staple crops that together cover about 670 Mha of har-
 117 vested area and 580 Mha of physical cropland: wheat, rice, maize, soybean
 118 and potato. Harvested areas (rainfed and irrigated) were taken from the
 119 SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1°
 120 cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn
 121 without replacement with probability proportional to harvested area. An un-
 122 weighted average over the sample therefore approximates an area-weighted
 123 average over global production. The 400 crop–site pairs lie in 381 distinct
 124 cells in 50 countries (Fig. 2). Sowing and maturity dates came from the
 125 GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat
 126 we used the winter-wheat calendar unless the coldest-month mean temper-
 127 ature was below −8 °C, in which case the spring-wheat calendar was used.
 128 Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,
 129 2018): global and diffuse shortwave irradiance, maximum, minimum and
 130 dew-point temperature, precipitation and wind speed at 2 m. Each site con-
 131 tributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days
 132 after sowing, which gives 7600 site-seasons.

133 Climate change was represented by the global warming levels +1.5, +2
 134 and +3 °C relative to 2001–2020. Daily maximum, minimum and dew-point
 135 temperatures were shifted uniformly, which keeps relative humidity approx-
 136 imately constant and raises the vapour-pressure deficit. The atmospheric
 137 CO₂ concentration was set to 390, 450, 500 and 600 ppm for the baseline and
 138 the three warming levels. Precipitation and irradiance were left unchanged.

139 This delta approach isolates the thermal signal of warming; its limitations
140 are discussed in Section 4.5.

141 2.2. A differentiable agrivoltaic simulator

142 The simulator advances all site-seasons in parallel with a daily time step
143 for the crop and an hourly time step for radiation and PV (Fig. 1). All equa-
144 tions are implemented as PyTorch tensor operations (Paszke et al., 2019), so
145 that yields and electricity are differentiable with respect to the design and
146 control variables.

147 *Hourly irradiance.* Daily global and diffuse irradiance were disaggregated
148 into hourly values with the Collares-Pereira and Rabl (1979) and Liu and
149 Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance
150 on the horizontal is the difference between the two. Solar position follows
151 the standard declination and hour-angle equations.

152 *Array geometry and PV electricity.* We consider elevated horizontal single-
153 axis trackers with north-south axes. The axis height is 1.25 module widths,
154 e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The ro-
155 tation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width
156 divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this
157 keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the
158 rows when the modules are horizontal, which is the stow position during field
159 operations. Let ψ be the solar zenith angle projected onto the east-west ver-
160 tical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,
161 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ oth-
162 erwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on
163 to the sun, which minimises the shadow on the crop. Given a light-sharing
164 level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

165 Plane-of-array irradiance combines three terms: beam irradiance on the mod-
166 ule, reduced by the fraction of the module shaded by the neighbouring row,
167 $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected
168 light (albedo 0.2). The cell temperature follows the NOCT model (45 °C),
169 with an hourly air temperature interpolated sinusoidally between the daily
170 extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with
171 $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature

172 losses and system losses. The installed capacity is 2.1 g MWp ha^{-1} . A con-
 173 ventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the
 174 energy reference E_{PV} .

175 *Validation of the PV model.* We compared the simulated specific yield of an
 176 isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2
 177 (European Commission, Joint Research Centre, 2022), for a horizontal north-
 178 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29
 179 returned data). Across sites, the simulated and PVGIS specific yields were
 180 correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean
 181 absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and
 182 with the different irradiance databases (NASA POWER vs. SARA5, ERA5
 183 and NSRDB). Electricity estimates are therefore, if anything, conservative.

184 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction
 185 of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The
 186 fraction of sky diffuse reaching the ground is a two-dimensional isotropic
 187 sky-view factor. We pre-computed it for 25 values of g and 13 rotation
 188 angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly
 189 (differentiably). The daily ratio s of the irradiance at the crop to that of the
 190 open field defines the crop’s shading. The canopy maximum temperature is
 191 lowered by $\kappa(1 - s)$.

192 κ is uncertain. Field studies under PV modules report small reductions
 193 of daytime crop or canopy temperature and little change in air temperature
 194 (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds
 195 calibrated on air temperature do not transfer directly to canopy temperature
 196 (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate
 197 $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$
 198 for all methods (Section 3.2). Nocturnal warming under the modules is not
 199 represented. The reference evapotranspiration under the modules, ET_0 , is re-
 200 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
 201 from the reduced radiation and canopy temperature. Soil water conserva-
 202 tion under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018)
 203 therefore emerges from the model, but rain redistribution by the modules
 204 (Marrou et al., 2013a) is ignored.

205 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 206 with the species parameters of its Table 1a (Table 1). Daily biomass growth

207 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

208 where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as
209 follows.

- 210 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
211 only on thermal time, with senescence accelerated by heat and drought
212 stress. Shade therefore reduces the daily biomass increment but not
213 leaf-area development.
- 214 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
215 treme temperature T_{ext} of the canopy maximum temperature.
- 216 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for
217 drought (Woli et al., 2012) is computed from a daily soil-water balance
218 with curve-number runoff, drainage and root water uptake. Irrigated
219 fields have $f_{\text{Water}} = 1$.

220 SIMPLE has a fixed harvest index and therefore ignores the high sen-
221 sitivity of grain number to shading around flowering (Fischer, 1985; An-
222 drade et al., 1999). It also ignores the differences in shade tolerance be-
223 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
224 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
225 reals and tuber crops initially lost less than proportionally; shading trials
226 also show marked effects of shade on maize development (Ramos-Fuentes
227 et al., 2023). We added a stage-dependent penalty. During a critical window
228 ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
229 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
230 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
231 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
232 0.5, for which no specific curve was available) and lowest for potato (0.2).
233 These values are assumptions that reproduce this ranking; they are not cal-
234 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
235 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
236 ciency of shaded canopies under more diffuse light. The penalty, conversely,
237 makes shading around flowering more damaging than proportional. The net
238 effect is closer to the field evidence than either assumption alone.

239 The thermal-time requirement to maturity was calibrated for every site as
240 the mean thermal time between the GGCM sowing and maturity dates under

the 2001–2020 climate, and the canopy parameters were scaled accordingly. The cultivar is thus adapted to the present climate, and warming shortens the season. Growth stops smoothly once the thermal-time requirement is met. Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site is represented by a rainfed and an irrigated sub-field in proportion to its SPAM irrigated fraction. Each sub-field carries its own array, so both yield *and* electricity are area-weighted over the two sub-fields.

2.3. SALS: simulation-trained adaptive light sharing

SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a multilayer perceptron with two hidden layers of 64 units and a bounded-logit output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs are the information available on day t :

- the crop and soil state: relative thermal time TT/T_{sum} , radiation interception f_{Solar} , relative soil water and the previous day’s ARID;
- a *forecast* of the day’s maximum temperature (relative to T_{heat} and T_{opt}), irradiance and ET_0 ;
- static information: the irrigation status of the sub-field, the array’s GCR g , latitude, day of season and the crop.

Forecasts are perturbed with random errors, independently every day: Gaussian with a standard deviation of 2°C for maximum temperature, and multiplicative with standard deviations of 20% for irradiance and 15% for ET_0 . The physics always uses the true weather. The same forecast errors are applied to the rule-based controllers. The controller is conditioned on g . During training, each season is simulated with a GCR close to the density that is sustainable at its training site. This is the largest value on a coarse grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention of the site’s training seasons at or above ρ , plus a uniform perturbation in $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to control the densities that design selection will later pick. Outside the cropping season the trackers backtrack for maximum electricity.

For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional

274 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

275 which combines an exact (hinge) and a quadratic penalty for the food-security
 276 floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the
 277 exact gradient of Eq. (3) is obtained by back-propagation through the 365
 278 daily steps of every season (back-propagation through time). This is the key
 279 difference from reinforcement learning with black-box simulators (Gautron
 280 et al., 2022), which estimates gradients from sampled returns.

281 Each minibatch contains 512 site-seasons, each simulated as a rainfed and
 282 an irrigated sub-field, drawn from one of the four climates in turn. Seasons
 283 with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training
 284 and evaluation. The network is fine-tuned through the simulator with Adam
 285 (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm
 286 clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in
 287 a numerically stable form, so that gradients remain finite even under extreme
 288 heat.

289 *Initialisation.* Trained from random weights, the controller converged to
 290 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
 291 the seasonal-sharing rule, because the gradient of electricity with respect to u
 292 is small and of the same sign for most days of a season. We therefore initialise
 293 the network by behaviour cloning of the phenology rule of Section 2.4: binary
 294 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
 295 1.7 million day-states visited by the rule in the training seasons, with 1500
 296 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
 297 policy away from this initialisation wherever doing so improves the objective.
 298 Consequently, the gain of SALS over the phenology rule measures what fine-
 299 tuning through the simulator adds to that rule, not what can be learned
 300 without it.

301 2.4. Deployable design selection and baselines

302 A real array has one fixed density, which must be chosen before it is built.
 303 Every method, SALS included, therefore uses the same *history-based design*
 304 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
 305 for which the method’s mean yield retention over the site’s 2001–2014 sea-
 306 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
 307 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

308 at least 80% of the historical seasons. The historical seasons of the test sites
309 are never used for training, and the weather of the evaluation years is never
310 used for design. We compare four controllers:

- 311 • *AV static*: backtracking trackers without light sharing ($u = 0$);
- 312 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
313 cropping season;
- 314 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
315 growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress
316 forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
317 track the sun and shade the crop;
- 318 • *SALS*: the learned controller.

319 The open field and a conventional PV plant without crops ($\text{LER} = 1$ each)
320 are the references. Because the design rule is identical for all controllers, dif-
321 ferences between methods reflect the quality of control and the array density
322 that each controller can sustain under the food-security floor. We also com-
323 pare the controllers at identical GCR over the whole grid (matched-density
324 frontiers).

325 2.5. Shade-response uncertainty: gradient calibration, robust controllers and 326 design

327 The crop’s response to shade is the least certain part of the simulator,
328 so we treat it as an unknown with a distribution. Two parameters per crop
329 enter the simulator: a radiation-use compensation c (growth is multiplied by
330 $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop)
331 and a multiplier h on the flowering-window harvest-index penalty. Because
332 the simulator is differentiable, both can be fitted by gradient descent through
333 whole simulated seasons to published yield–shading observations. The obser-
334 vations are the relative yields at 20% and 40% shading read from the curves
335 of Laub et al. (2022) for the five crops, the field trials on rice, wheat, potato
336 and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al.,
337 2023) (each season of each trial as a separate observation, with an obser-
338 vation error of 0.06–0.08) and the unaffected harvest index of shaded maize
339 (Ramos-Fuentes et al., 2023). Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and $h \sim \mathcal{N}(1, 0.5^2)$,
340 truncated at zero. We fitted 12 ensemble members by randomised maximum
341 a posteriori estimation: every member sees the observations perturbed by
342 their errors and a prior draw perturbed by its width, and is optimised with
343 Adam (40 steps, 40 training-block seasons per crop and member, shading of
344 the static arrays cached). The ensemble approximates the posterior of the
345 shade response and is used in three ways.

346 *Robust and adaptive controllers.* SALS-robust has the architecture and
347 training of SALS (Section 2.3), but every training season is simulated with the
348 response of a randomly drawn ensemble member that the controller does not
349 observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible
350 densities differ between members. SALS-adaptive additionally observes the
351 realised biomass of the crop under the modules relative to an open reference
352 plot, a quantity that can be measured on the farm by remote sensing or by
353 a reference plot. A feedback rule that shares light whenever this ratio falls
354 below a target serves as the hand-written counterpart; its target (1.0) was
355 chosen on training-block seasons under the ensemble.

356 *Uncertainty-aware design.* The design rule of Section 2.4 chooses the
357 largest GCR whose historical retention meets the floor. Under shade-response
358 uncertainty we choose the largest GCR that meets the floor in at least 75%
359 of five ensemble members, and compare it with a design under the ensemble
360 mean and with an oracle design under the true response. Evaluation of this
361 rule is leave-one-out: the truth is the held-out member and the design uses
362 the other four.

363 *Stress tests.* Controllers and rules are also evaluated under four fixed
 364 responses, the proportional response, the response calibrated to the meta-
 365 analysis, the response fitted to the field trials and a harsh response (propor-
 366 tional growth and a 1.5 times larger harvest-index penalty). The proportional
 367 and harsh responses lie outside the range of the ensemble.

368 2.6. Evaluation protocol, statistics and upscaling

369 Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to train-
 370 ing (70%) or test (30%), so that no test site has a training site in the same
 371 block. Training uses the 2001–2014 seasons of the training sites. Design
 372 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the
 373 2015–2019 seasons of the test sites. The same site-seasons are analysed in all
 374 four climates.

375 We report the following metrics:

- 376 • the mean electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$) and the relative electricity e ;
- 377 • the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- 378 • season-level compliance (share of seasons with $r \geq \rho$) and site-level
 379 compliance (share of sites whose mean test-season retention is $\geq \rho$);
- 380 • the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- 381 • the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

382 Statistical comparisons use site means as the unit of analysis. We report
 383 paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confi-
 384 dence intervals of the mean difference. SALS was trained with three random
 385 seeds on the main split, and with one seed on each of two further spatial
 386 splits.

387 For the global estimate, every site is evaluated in the split in which it is
 388 held out. The mean electricity and yield per hectare of each crop are scaled
 389 to the *physical* area of that crop in SPAM 2010, which avoids double counting
 390 multiple-cropped land (Waha et al., 2020). We report the installed capacity
 391 (TWp), the capacity factor and a supply curve.

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in $^{\circ}\text{C}$, thermal times in $^{\circ}\text{C d}$, RUE in gMJ^{-1} ; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

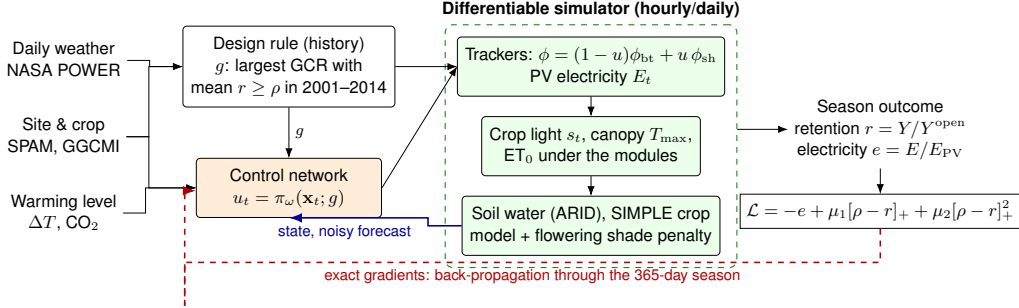


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out sites of the main split (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By construction all methods keep the mean yield retention close to 90% (0.904–0.917), so differences in electricity are differences in the density that each controller can sustain under this floor.

Static arrays are the weakest. Without light sharing, the floor limits the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$, 16% of a conventional PV plant per hectare (LER 1.07). Rotating the

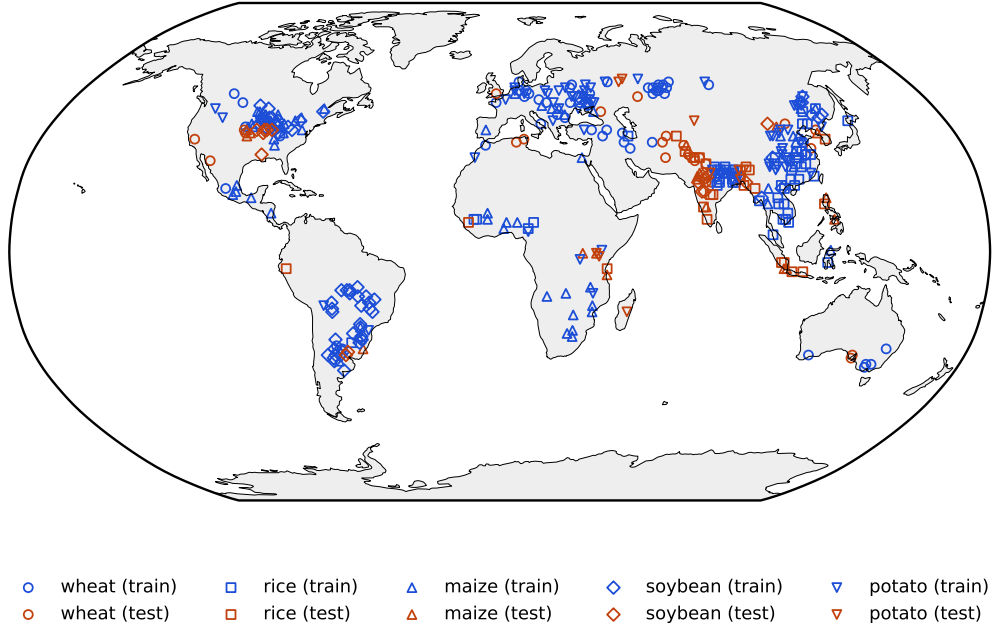


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

modules to let light through during the cropping season (seasonal sharing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule that shares light only while the crop is growing (phenology rule) reaches $465 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.24). An expert rule that additionally shades the crop on forecast heat or drought days (stress rule) is worse than the phenology rule (LER 1.18): its density is limited to 13% because it reduces light at times when the floor is most binding.

The learned controller is best by a modest, consistent margin. SALS sustains a density of 17% (similar to the phenology rule), produces $510 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ and reaches an LER of 1.270 at the same retention. Compared with the phenology rule this is +0.038 in LER (95% CI over sites 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Table A.9). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows the same picture at matched densities: the learned policy lies on or slightly

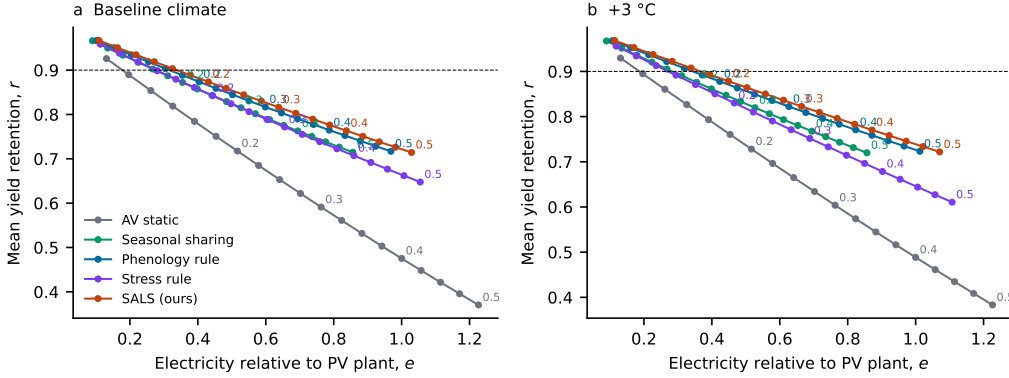


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

above the phenology rule over the whole density range, both lie well above seasonal sharing and the stress rule, and static arrays are far below.

What the floor does and does not guarantee. Enforcing the floor on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at +3 °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split: 0.034 and 0.032). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, includ-

ing the historical design step, with $\kappa = 0, 1$ and 2°C . Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at 0.905 by the design rule and the ranking is unchanged. SALS exceeds the phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate (0.022, 0.032 and 0.051 at $+3^\circ\text{C}$).

Which ingredients matter. Two ablations test the design of SALS at the same training budget. Masking the stress inputs (soil water, ARID, and the maximum temperature, irradiance and ET₀ forecasts) leaves the LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297 at $+3^\circ\text{C}$). Training only on the baseline climate gives 1.268 and 1.291. The improvement over the rules therefore does not depend on stress-aware inputs or on training in warmed climates.

3.3. Floor conditioning, chance-constrained training and forecast errors

One network for all floors. Regulations and farmers set different floors. The network of the main experiments was trained at $\rho = 0.9$ only. We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical design step for each floor (Table 3). Its LER exceeds that of the phenology rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037, 0.032 and 0.033 at $+3^\circ\text{C}$), so the benefit carries over to other floors without retraining. A network that received the floor as an additional input and was trained over all four floors did not improve on it (differences in LER ≤ 0.001). Because the design step already adapts the density to the floor, conditioning the controller on the floor is not needed.

Chance-constrained training. We also trained a network whose objective penalises a batch share of seasons below the floor that exceeds 20%. It did not improve the tail. With the mean-based design step its LER was 1.263 (baseline) and 1.290 ($+3^\circ\text{C}$) against 1.270 and 1.297, and with chance-constrained design it reached season-level compliance of 92.6% against 92.0% and an LER of 1.198 against 1.205. The season-level guarantee is obtained by the design step (Section 3.1), not by the training objective.

Forecast errors. We raised the forecast errors seen by the controllers to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and of the phenology rule was identical at all four error levels to three decimals (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and 1.416). Neither controller uses the weather forecast: both rely on the crop’s thermal time, which is observable from on-farm temperature records. The

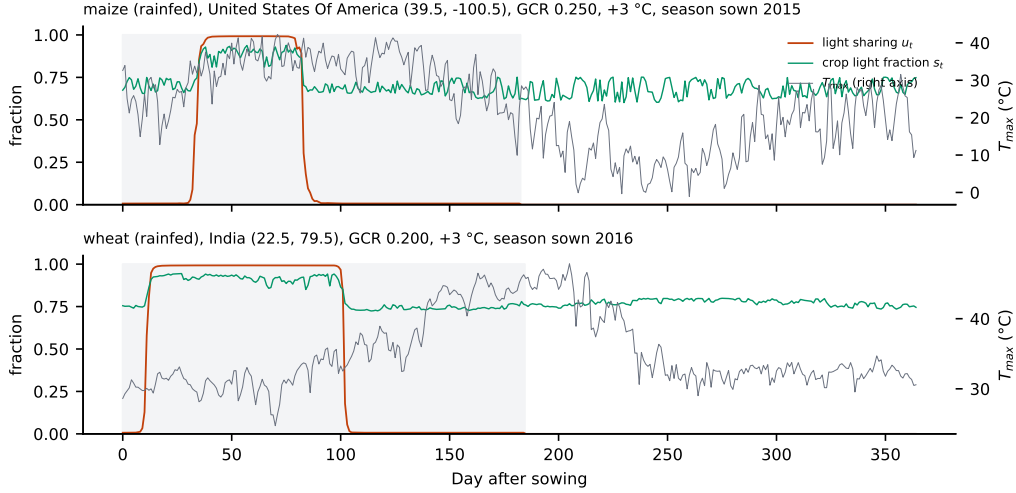


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

477 stress rule, which depends on the forecast of maximum temperature, degrades
 478 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
 479 its LER from 1.432 to 1.418 as the errors triple.

480 3.4. What the learned policy does

481 Averaged over the rainfed test seasons of the main split at a density of 0.3,
 482 the light-sharing level of SALS is a smooth function of relative thermal time.
 483 It is close to zero in the first 10% of the thermal time required for maturity,
 484 rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and
 485 falls through 0.58 at 80–90% to about zero at maturity. The plateau covers
 486 the window in which our crop model penalises shading through the harvest
 487 index. The ramps at both ends let the modules produce electricity when
 488 additional light is worth least to the crop. Fig. 4 shows two example seasons.
 489 The phenology rule, which is also the initialisation of the network, shares
 490 light during the whole growing period. Fine-tuning through the simulator
 491 found that sharing can be reduced at the edges of the season: SALS gains its
 492 advantage by yielding light when it matters and collecting electricity when
 493 it matters less.

494 The policy also shares less light on days with a forecast heat stress: the
 495 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 (+3 °C)
 496 on other days, and 58–63% of sites show a lower level on heat days. This
 497 cannot be read as a stress response, however. Heat days occur late in the
 498 season, when the phenological ramp is already falling, and a policy without
 499 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).
 500 The ablation above confirms that stress information does not improve the
 501 result.

502 3.5. Crops and regions

503 The benefit of light sharing differs between crops (baseline climate, SALS).
 504 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
 505 577 MWh ha⁻¹ yr⁻¹; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
 506 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
 507 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
 508 1.14–1.21 for the others).

509 The floor is not met evenly. Production-weighted retention is 0.888 for
 510 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
 511 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
 512 combine the highest density (GCR 0.41) and electricity (1090 MWh ha⁻¹ yr⁻¹)
 513 with a production-weighted retention of only 0.862. Africa and the Middle
 514 East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention
 515 of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902
 516 (67% of sites) at a density of 0.14. A regulatory floor enforced per site would
 517 therefore need a region- and crop-specific design step, not a global one.

518 3.6. Model behaviour and validation

519 **PV output.** The simulated specific yield of an isolated tracker row agrees
 520 with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29
 521 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation
 522 of 5.5% (Section 2.2).

523 **Shade response.** Fig. 5 shows the simulated relative yield of rainfed
 524 crops under static arrays as a function of the season-mean shading of the crop.
 525 Yield declines close to proportionally with shading: 84–89% relative yield at
 526 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat
 527 loses most and potato and maize least. This response is steeper than the *less*
 528 *than proportional* losses that Laub et al. (2022) report for small shade levels
 529 of C3 cereals and tubers, and it does not reproduce their finding that maize

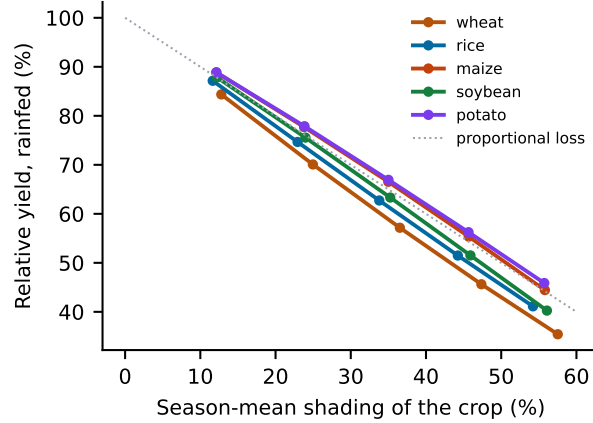


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

Comparison with field trials. We fitted one radiation-use compensation coefficient per crop to the yield–shading curves of Laub et al. (2022) using training-block sites only (Methods, calibration script in the repository). The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the proportional model already lies within about 0.02 of the C3-cereal and tuber curves, and it cannot lower the maize and legume responses below the proportional case (root-mean-square error 0.16 for maize and 0.08 for soybean). Against independent field trials (Table 4) the model underestimates retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al. (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 5 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the

field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains reported in this paper are thus lower bounds with respect to these trials and not upper bounds.

3.7. Learning under shade-response uncertainty

Calibration. Gradient descent through the simulator fitted the shade response of all five crops to the meta-analysis and the field trials with small residual spread in the yield curves (relative yield at 23–25% shading 0.83 ± 0.02 – 0.03 for rice and maize, 0.84 ± 0.03 for wheat, 0.86 ± 0.03 for potato and 0.72 ± 0.04 for soybean), whereas some parameters remain uncertain (Table 6). The data identify two features: the harvest index of maize is hardly affected by shade (multiplier 0.28 ± 0.14 , $c \approx 0$), as observed by Ramos-Fuentes et al. (2023), and soybean is more shade sensitive than the cereals (compensation 0.12, harvest-index multiplier 1.38), as in Laub et al. (2022). The compensation of wheat is the least certain (0.68 ± 0.34).

Learned controllers. Under every response the learned controller stays better than the best hand-written rule, but only slightly (Table 7): the LER gain of SALS over the phenology rule is +0.014, +0.015, +0.033 and +0.010 under the proportional, calibrated, field-fitted and harsh responses. Training on the ensemble did not help: SALS-robust differs from SALS, which was trained on the proportional response only, by at most 0.005 in LER, and SALS-adaptive, which sees the measured biomass ratio, by at most 0.008. The feedback rule is worse than the phenology rule by 0.025–0.064 in LER. A plausible reason is that the biomass ratio does not reveal the harvest-index loss from shading around flowering, so the information that the controller could use to adapt is not in the measurement.

Design under uncertainty. The larger risk is in the design step (Table 8). When the density is chosen under the ensemble mean and the true response is a different plausible one, the share of sites whose mean retention reaches the floor falls to 58% (phenology rule) and 55% (SALS), against 76% and 74% for an oracle design. The uncertainty-aware design restores the oracle level of site compliance (76% and 74%) at a cost of 0.029 and 0.023 in LER relative to the ensemble-mean design. For the proportional and harsh responses, which lie outside the ensemble, the same design raises site compliance of the phenology rule from 29% to 33% and from 14% to 24% but does

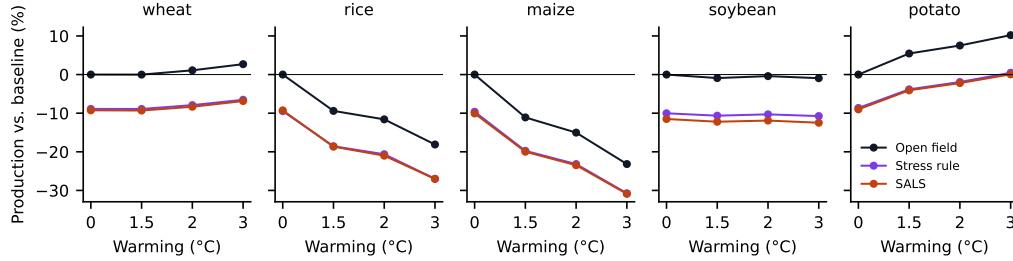


Figure 6: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

not restore it, so the guarantee holds only within the range of responses that the data support.

3.8. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 6) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize −31%, rice −27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha^{−1} yr^{−1}). The margin of SALS over the rules does not grow with warming (Table A.9).

3.9. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop-site pairs, in at least one split), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr^{−1} under the baseline climate and 16.0 PWh yr^{−1} at +3 °C, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr^{−1}, and static arrays 6.4 and 6.9 PWh yr^{−1}. For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

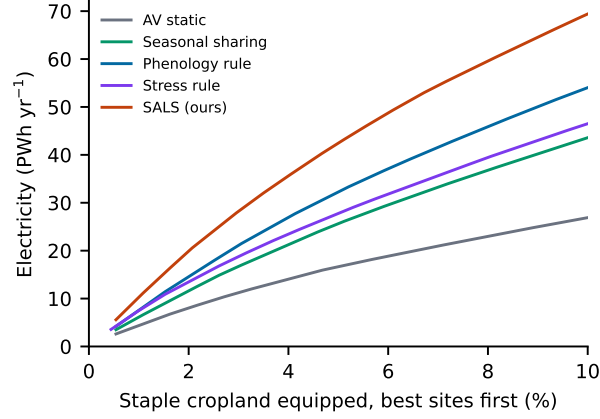


Figure 7: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at +3 °C), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 7): the best 2% of the staple cropland would supply 16 PWh yr⁻¹ and the best 5% 40 PWh yr⁻¹ with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

4. Discussion

4.1. What the results establish

Four findings stand on consistent evidence. First, sharing light with the crop is the main lever on the electricity that tracker-based agrivoltaic arrays can produce under a yield floor. With static trackers the floor limits arrays to about 6% ground coverage and an LER of 1.07. Any light-sharing control

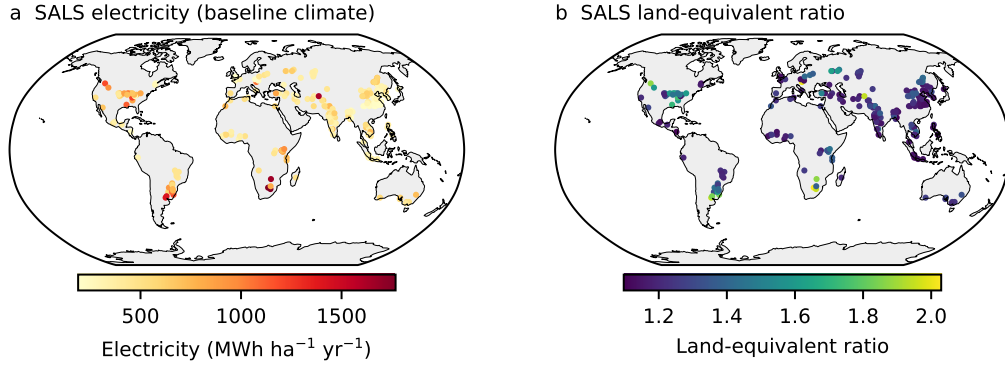


Figure 8: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

631 allows about three times the density, and the LER rises to 1.18–1.24 for hand-
 632 written rules. Second, the best simple rule follows crop phenology. It shares
 633 light across the middle of the season, around flowering and grain filling, and
 634 produces electricity at the beginning and the end. It needs only the crop’s
 635 thermal time, which is observable on the farm, and it is insensitive to forecast
 636 errors, whereas the rule that reacts to heat and drought stress is worse in
 637 every climate and degrades with forecast error. Third, the qualitative result
 638 does not depend on the weakest part of the model. The simulated shade
 639 response is conservative against the field trials we could use (Section 3.6);
 640 calibrating it to a meta-analysis or fitting it to the trials raises the allowed
 641 density, and the phenology rule stays the best method, with a gain over static
 642 arrays of 0.17, 0.19 and 0.30 LER under the conservative, calibrated and field-
 643 fitted responses. The absolute gains reported are therefore lower bounds with
 644 respect to those trials, not upper bounds, but the trials are few, cover four
 645 crops and one or two seasons each, and disagree with the meta-analysis for
 646 maize. Fourth, a controller learned through the differentiable simulator is
 647 better than the best rule we could write, by 0.03–0.05 in LER and about
 648 10% in electricity at equal retention, reproducibly over training seeds (range
 649 0.002 in LER), three spatial splits, three canopy-cooling assumptions and four
 650 climates, and at 84–86% of the held-out sites. This gain is small compared
 651 with the gap between static and dynamic control.

652 *4.2. What role does artificial intelligence play?*

653 We tested three ways in which learning could matter in this problem,
654 and the answers differ. The first is calibration. Because the simulator is
655 differentiable, the crop’s response to shade could be fitted by gradient de-
656 scent through whole seasons to a meta-analysis and four field trials, giving
657 an ensemble of plausible responses that reconciles them (Section 3.7). The fit
658 identified two properties that the sources suggest separately, an unaffected
659 harvest index of maize and a higher shade sensitivity of soybean, and it
660 supplies the uncertainty that the next step needs. This is the part where
661 differentiable programming clearly paid off: a hand calibration of ten param-
662 eters against twenty-one observations would have been slower and would not
663 have produced an ensemble.

664 The second is control. Here learning contributes little. The learned con-
665 troller adds 0.01–0.03 LER over the best hand-written rule, which follows
666 crop phenology, under every response we tested, and a reader who needs a
667 simple, auditable controller can use that rule and obtain most of the electric-
668 ity gain over static arrays. Neither of the two extensions that we expected
669 to matter under uncertainty did: training on an ensemble of responses did
670 not change the policy, and giving the controller a farm-measurable signal,
671 the biomass of the crop relative to a reference plot, did not either, nor did
672 a feedback rule built on the same signal. The learned policy also has to be
673 initialised from the phenology rule (Section 4.5). We therefore do not claim
674 that learned control is what makes agrivoltaics work. We claim that, in this
675 model, the schedule that a learned policy converges to is a smooth function
676 of crop phenology, which is what a rule can already express.

677 The third is design under uncertainty, and this is where the calibrated
678 ensemble changes decisions. A density chosen under the ensemble mean
679 meets the floor at 55–58% of sites if the true response is another plausible
680 member, against 74–76% with an oracle design; choosing the density so that
681 the floor is met in at least 75% of plausible responses restores the oracle
682 level at a cost of 0.02–0.03 LER. The bottleneck of tracker-based agrivoltaics
683 under a food-security floor is therefore not the controller but the quantified
684 uncertainty of the crop response that enters the design, and it is data on that
685 response, not larger networks, that would improve the result. The guarantee
686 holds only within the range of responses that the data support: it improved
687 but did not restore compliance under the proportional and harsh responses,
688 which lie outside the ensemble.

689 *4.3. What the results do not establish*

690 The early hypothesis that stress-aware control, that is, shading the crop
691 on heat and drought days, would be a main source of benefit under warming
692 was not supported. The stress rule performs worse than the phenology rule
693 in every climate, and a learned controller without stress inputs performs as
694 well as one with them. A plausible reason is that in our crop model the yield
695 gain from shading on a hot or dry day is smaller than the loss of light on
696 that day, so stress information does not help. Whether this holds for crop
697 models with more detailed canopy-temperature and heat-stress responses is
698 an open question. Agrivoltaics does not protect staple production against
699 warming in our simulations: light-sharing systems produce about 90% of the
700 open-field yield in every climate and follow the same decline in maize and
701 rice (Fig. 6).

702 *4.4. Food security in practice*

703 Holding the historical mean retention at 90% does not hold it in each
704 season or at each site. 70% of SALS seasons meet the floor, the lower tail is
705 slightly heavier than for the rules, and production-weighted retention is below
706 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-
707 constrained design raises season-level compliance to about 92% at a cost of
708 0.065 in LER, and the ranking of the controllers is preserved. Whichever for-
709 mulation is chosen, a per-site regulatory floor would need a region- and crop-
710 specific design step. The upscaled potential (Section 3.9) shows the trade-off
711 in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply
712 need about 10 TWp of modules, several times the present global fleet, and
713 the best sites are rainfed maize and soybean areas where production-weighted
714 retention is lowest.

715 *4.5. Limitations*

- 716 • *Simulation only.* All results come from a simulator. PV output was
717 benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop
718 response to shade was only compared with a handful of field trials and
719 one meta-analysis (Section 3.6). The simulated response is close to
720 proportional, is conservative with respect to the trials (rice, wheat,
721 potato, maize) and does not reproduce the stronger losses of maize
722 and legumes in Laub et al. (2022). The harvest-index sensitivities h
723 are assumptions, and the sensitivity analysis uses rule controllers on a
724 subsample of 22 sites.

- 725 • *Canopy cooling.* The effect of shade on canopy temperature was repre-
726 sented by one coefficient κ , applied to a heat-stress function that was
727 calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018).
728 The ranking of the controllers is robust to κ between 0 and 2 °C, but
729 the absolute electricity changes by about $\pm 8\%$. Nocturnal warming
730 under modules and rain redistribution are not modelled.
- 731 • *Crop model.* SIMPLE has a fixed harvest index and linear radiation-
732 use efficiency. More detailed models with grain-number formation and
733 canopy energy balance may favour different schedules.
- 734 • *Climate scenarios.* Warming was applied as uniform temperature shifts
735 with fixed precipitation, radiation and relative humidity, and with un-
736 changed cultivars and sowing dates. It is not a bias-adjusted climate
737 projection.
- 738 • *Initialisation and state information.* The network is initialised from the
739 phenology rule, because training from random weights collapsed to full
740 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,
741 and we cannot say whether a better optimiser or a richer policy class
742 would find larger gains. The controller uses the simulated state of the
743 crop and soil (thermal time, canopy interception, soil water) without
744 measurement error, and the forecast errors we tested apply to weather
745 only.
- 746 • *Economics and deployment.* There is no cost, market or grid analysis.
747 The value of midday electricity, curtailment, storage, structural cost
748 and farmer acceptance (Trommsdorff et al., 2021) will determine the
749 realisable share. The average row pitch at the chosen densities (about
750 13 m for 2.2-m modules) leaves room for machinery, but the densest
751 designs (GCR up to 0.5) have a pitch of only 4.4 m.

752 5. Conclusions

753 Light sharing, the daily choice between maximum electricity and letting
754 light through to the crop, is the main lever of tracker-based agrivoltaics un-
755 der a food-security floor. In a differentiable simulator of trackers, PV output,
756 crop light, microclimate and crop growth, evaluated fairly on 400 crop-site

757 pairs with the array density of each site chosen from its own history, light-
 758 sharing rules allowed about three times the module density of static arrays
 759 at a mean yield retention of 90%. They raised the land-equivalent ratio
 760 from 1.07 to 1.18–1.24, and a rule that follows crop phenology and needs no
 761 weather forecast was the best of them. The simulated shade response is con-
 762 servative against the field trials we could use, so these gains are lower bounds,
 763 and calibrating the response raised them without changing the ranking of
 764 methods. A learned controller added a further 0.01–0.05 LER, by following a
 765 smooth phenology-dependent schedule that a rule can express; training it on
 766 an ensemble of shade responses or giving it a farm-measurable biomass signal
 767 added nothing. The part where learning through the differentiable simulator
 768 paid off was the calibration of the shade response, which gave an ensemble of
 769 plausible responses; designing the array density so that the floor holds in 75%
 770 of them restored the compliance of an oracle design at a cost of 0.02–0.03
 771 LER. The yield floor is met on average but not in every season or region,
 772 and the electricity potential of 5% of staple cropland (about 14 PWh yr⁻¹)
 773 requires about 10 TWp of modules. Future work should validate the crop
 774 response to shade under trackers with more field data, notably for maize and
 775 legumes where the meta-analysis and the trials disagree, extend the crop
 776 model to grain-number formation and canopy energy balance, add costs and
 777 grid constraints, and pursue controllers that guarantee the food-security floor
 778 per site and per season.

779 **CRedit authorship contribution statement**

780 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal
 781 analysis, Investigation, Data curation, Visualization, Writing – original draft,
 782 Writing – review & editing.

783 **Declaration of competing interest**

784 The author has no known competing financial interests or personal re-
 785 lationships that could have appeared to influence the work reported in this
 786 paper.

787 **Data and code availability**

788 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
 789 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA

790 POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters
791 (Zhao et al., 2019). The code of the simulator, of SALS and of all experi-
792 ments, together with the site list, is available from the author’s public reposi-
793 tory (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

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797 Appendix A. Statistical comparison

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid $0.05, 0.075, \dots, 0.5$ at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: mean of reported seasons; for wheat and potato the two-year range is -18.7 to $+2.7\%$ and -18.2 to $+11\%$.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 5: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 6: Shade-response parameters calibrated by gradient descent through the simulator (ensemble of 12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier on the flowering-window harvest-index penalty h (1 = assumed value).

Crop	c	h multiplier
Wheat	0.68 ± 0.34	0.88 ± 0.53
Rice	0.38 ± 0.27	0.97 ± 0.33
Maize	0.01 ± 0.01	0.28 ± 0.14
Soybean	0.12 ± 0.17	1.38 ± 0.33
Potato	0.38 ± 0.19	0.97 ± 0.48

Table 7: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out site, 21 sites). The proportional (conservative) and harsh responses lie outside the range of the calibrated ensemble.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule	1.164	1.190	1.456	1.122
SALS (trained on proportional response)	1.178	1.209	1.489	1.132
SALS-robust (ensemble-trained)	1.179	1.204	1.491	1.127
SALS-adaptive (ensemble, biomass feedback)	1.176	1.202	1.495	1.124

Table 8: Array design under shade-response uncertainty. Truth: each of five ensemble members in turn. Designs: oracle (density chosen under the truth), ensemble mean (density chosen under the mean response) and uncertainty-aware (density chosen so that the floor is met in at least 75% of the *other* four members). Means over the five held-out members; retention is the mean season retention, site compliance the share of sites whose mean retention is ≥ 0.9 .

Design	Method	GCR	Retention	Seasons ≥ 0.9 (%)	Sites ≥ 0.9 (%)	LER
oracle	Phenology rule	0.198	0.911	75	76	1.296
oracle	SALS	0.193	0.911	75	74	1.314
ensemble mean	Phenology rule	0.206	0.906	59	58	1.305
ensemble mean	SALS	0.196	0.907	61	55	1.318
uncertainty-aware	Phenology rule	0.184	0.916	77	76	1.276
uncertainty-aware	SALS	0.181	0.916	77	74	1.295

Table A.9: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92